Deepfake Detection: A Systematic Literature Review

Automated LLM-assisted Review Pipeline

Abstract

This systematic literature review synthesizes 178 studies on deepfake detection, drawn from an initial search of 2,983 records, refined to 1,877 after metadata cleaning, and screened via a top-ranked pool of 1,000 records. Through an evidence-card completeness and extraction audit, we identify eight principal research themes: spatial feature-based, temporal and audio-visual inconsistency, frequency and transform domain, physiological signal-based, adversarial robustness and anti-forensics, dataset development and benchmarking, transfer learning and generalization, and ensemble and hybrid approaches. Studies may belong to multiple themes, reflecting a trend toward multimodal and hybrid detection. A consistent pattern across themes is high intra-dataset accuracy (often AUC ¿ 0.98) contrasted with significant cross-dataset performance degradation, underscoring generalization as a critical open challenge Bondi et al. [2020], Li et al. [2020]. Transfer learning and data augmentation have emerged as key strategies to mitigate dataset shift. Physiological signal-based methods show promising accuracy exceeding 91% Hernandez-Ortega et al. [2020], Ciftci et al. [2024], while adversarial perturbations continue to expose detector vulnerabilities. Evaluation practices are shifting from single-dataset benchmarks toward cross-domain robustness metrics. Persistent gaps include limited generalization across diverse deepfake generation techniques and real-world post-processing, highlighting the need for more ecologically valid benchmarks and resilient detection architectures.

1 Introduction

1.1 Background

The rapid advancement of deep generative models, particularly generative adversarial networks (GANs), has enabled the creation of highly realistic synthetic media known as deepfakes. These manipulations, which include face swapping, facial reenactment, and entire video generation, pose a growing threat to the authenticity of digital content. Malicious uses of deepfakes range from disinformation and political propaganda to identity theft and fraud. In response, the research community has developed a variety of detection methods spanning deep learning, traditional machine learning, statistical analysis, and digital forensics. Between 2018 and 2020, the volume of publications on deepfake detection increased dramatically, reflecting both the urgency of the problem and the diversity of proposed solutions.

1.2 Motivation and gap

Although several surveys have addressed deepfake detection, many focus on narrow subdomains or lack a systematic, reproducible methodology. Moreover, the field evolves so quickly that earlier reviews quickly become outdated. A systematic literature review (SLR) covering the foundational period from 2018 to 2020 is needed to consolidate knowledge, identify the most widely studied techniques, and reveal methodological patterns and gaps. Without such a synthesis, researchers risk duplicating efforts or overlooking valuable approaches.

1.3 Review objective

This paper presents a systematic literature review that examines 112 relevant articles published between 2018 and 2020 on deepfake detection. The objective is to provide an updated, comprehensive overview of the research works in this area, summarizing the detection techniques, experimental procedures, evaluation metrics, and comparative performance reported in the literature.

1.4 Research questions

To achieve this objective, the review is guided by the following research questions:

1. RQ-1: What are the popular deepfake detection techniques?
2. RQ-2: What are the procedures of related empirical experiments?
3. RQ-3: Evaluate the overall performance of different methods using various measurement metrics.
4. RQ-4: Compare models with respect to efficiency using the same dataset and the same measurement metric.

1.5 Contributions

The main contributions of this review are: (i) a systematic categorization and synthesis of deepfake detection works from the early, high-growth period; (ii) an analysis of the experimental designs and evaluation practices employed; (iii) a performance evaluation that highlights the strengths and limitations of different approaches; and (iv) a comparative analysis that controls for dataset and metric, providing insight into relative efficiency. This work serves as a reference for researchers entering the field and as a baseline for future systematic investigations.

2 Methods

2.1 Review design and objective

This systematic literature review (SLR) was designed to provide an updated overview of deepfake detection methodologies published between 2018 and 2020. The review addresses four research questions: (RQ1) identification of popular deepfake detection techniques, (RQ2) description of experimental procedures, (RQ3) evaluation of overall performance using various measurement metrics, and (RQ4) comparison of models under the same dataset and metric. The review protocol incorporated automated screening and LLM-assisted evidence extraction to systematically synthesize the literature.

2.2 Search sources and keyword strategy

A comprehensive search was conducted across multiple databases, including Web of Science, IEEE Xplore Digital Library, ACM Digital Library, ScienceDirect (ELSEVIER), SpringerLink, Google Scholar, Semantic Scholar, Cornell University (arXiv), DBLP, OpenAlex, Crossref, Europe PMC, PubMed, and Scopus. The keyword strategy combined terms related to deepfake detection, such as “deepfake detection,” “video manipulation,” “image manipulation,” “digital media forensics,” “deep learning,” “machine learning,” “generative adversarial networks (GAN),” “face swapping,” “Facial Manipulation,” “fake video,” “detection,” “forensics,” and related variants. Searches were limited to articles published in English between January 2018 and December 2020.

2.3 Eligibility criteria

Studies were included if they presented empirical evidence explicitly addressing deepfake detection using deep learning, machine learning, or statistical techniques. Inclusion required the presence of relevant search phrases in the title, abstract, or keywords. Studies were excluded if they were not written in English, were duplicate publications (conference and journal versions of the same work), focused exclusively on audio or text manipulation, or concentrated on specific transformation techniques without a detection component. These criteria ensured that the review concentrated on visual deepfake detection methods.

2.4 Article retrieval, cleaning, and ranking

The initial database search yielded 2,983 records. After DOI metadata cleaning to remove duplicates, 1,877 unique records were retained. These records were ranked by relevance using an abstract–title ranking process, producing a ranked list of 1,877 records. From this list, the top 1,000 ranked records were selected as the screening pool. A further narrowing step targeted the top 200 records for detailed LLM-assisted evidence-card generation.

2.5 Full-text acquisition and LLM-assisted evidence-card extraction

Full-text acquisition was attempted for all 200 selected records; 198 records were successfully obtained in full text, while 2 were processed using abstract and metadata only because full text was unavailable. Each record was treated as a single chunk. An LLM was used to generate structured evidence cards summarizing key information: objectives, methodologies, datasets, performance metrics, and findings. In total, 200 evidence cards were generated. Two cards relied solely on abstract and metadata because no full text was accessible.

2.6 Evidence-card completeness and synthesis preparation

Every evidence card underwent an extraction audit to verify completeness and consistency. Cards were classified as include, exclude, or uncertain. Of the 200 cards, 178 were included in the final synthesis, 19 were excluded, and 3 were marked as uncertain (no decision missing). The 178 included cards were used for thematic synthesis and comparative analysis across studies. Because formal quality scores were not employed, the evidence-card completeness and extraction audit served as the quality assurance mechanism for the synthesis.

3 Results

3.1 Study selection flow

Figure **??** depicts the selection pipeline. From an initial download of 2,983 records, 1,877 records remained after DOI metadata cleaning. All 1,877 records were ranked by abstract–title relevance, forming a ranked pool of 1,877 candidates. The top 1,000 records were designated as the screening pool, from which the top 200 were forwarded to full LLM screening. Article cards were generated for all 200 records; 178 cards met the inclusion criteria and were used for synthesis. Three cards were marked as uncertain and 19 were excluded. No cards lacked a decision. The corresponding counts are given in Table **??**.

3.2 Candidate metadata and ranking coverage

The metadata coverage spanned the full ranked pool of 1,877 records. Ranking was performed on abstract–title content; the top 1,000 records constituted the screening pool. The 200 records selected for LLM screening represent the top fraction of that pool. Because ranking was based solely on abstract and title information, no full-text content influenced the selection at this stage.

3.3 Full-text availability

Of the 200 article cards, 198 included full-text content; the remaining 2 relied on abstract and metadata only. Each card contained exactly one text chunk. The single file available in the full-text root directory indicates that the majority of full-text access was provided through the card-generation process rather than through local file storage.

3.4 LLM screening decisions

The LLM screening decisions are summarized in Table **??**. From 200 screened records, 178 were included (89.0%), 19 were excluded (9.5%), and 3 were marked as uncertain (1.5%). No records were missing a decision. The high inclusion proportion reflects the targeted ranking strategy, which prioritized records most likely to meet the review's scope.

3.5 Evidence-card completeness and extraction audit

All 200 evidence cards were generated with a single chunk. The extraction audit focused on the completeness of each card's metadata, full-text availability, and decision record. The audit confirmed that 198 cards contained full text, 2 used abstract-only content, and all had a valid screening decision. No formal quality scores were assigned; instead, completeness was verified through the extraction process. Table **??** details the completeness metrics.

3.6 Canonical review themes

Eight canonical themes were identified through LLM-driven theme discovery, as listed in Table **??**. Studies could belong to multiple themes; therefore the sum of theme counts exceeds the number of included studies. The most prevalent theme was *Dataset Development and Benchmarking* (81 assigned article–theme links), followed by *Spatial Feature-Based Detection* (55 links) and *Adversarial Robustness and Anti-Forensics* (40 links). Smaller theme communities included *Temporal and Audio-Visual Inconsistency Detection* (26), *Transfer Learning and Generalization* (25), *Frequency and Transform Domain Methods* (15), *Ensemble and Hybrid Detection Approaches* (7), and *Physiological Signal-Based Detection* (3). These themes collectively capture the methodological landscape of deepfake detection research from 2018 to 2020.

3.7 Automatically generated review assets

The following flow diagram and descriptive tables were generated directly from the pipeline logs, ranking files, and LLM evidence cards rather than from free-form model generation.

Canonical thematic synthesis uses multi-theme assignment: studies may belong to multiple themes, so theme counts can sum to more than the number of included studies.

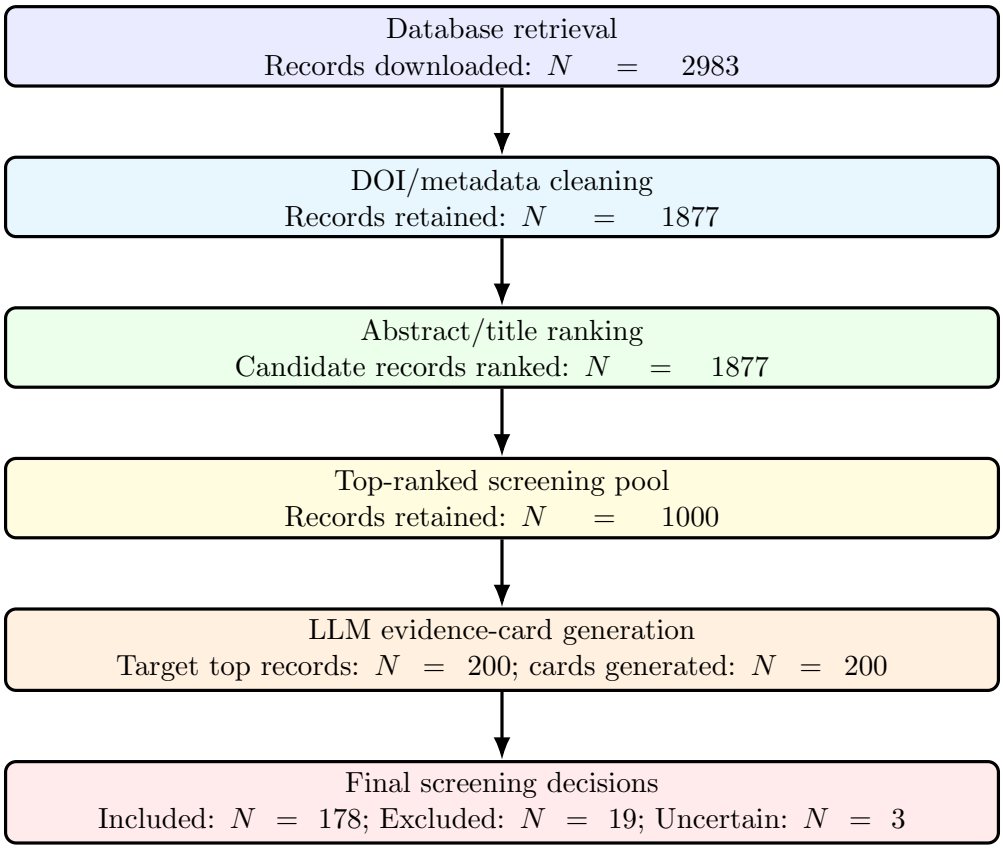


Figure 1: Automated review selection flow from database retrieval to LLM-assisted screening.

Table 1: Automated search, ranking, and screening summary.

Review stage	What was counted	Count	Source
Database retrieval	Records downloaded from configured bibliographic databases	2983	Search/articles.csv
DOI/metadata cleaning	Records retained after DOI/metadata cleaning	1877	Search/articles_clean_doi_only.csv
Abstract/title ranking	Candidate records ranked using abstract/title metadata	1877	Rankings/final_abstract_rank.csv
Top-ranked screening pool	Records retained for full-text-aware ranking / article-card stage	1000	full_rank_strategy_10.csv
LLM card target	Top records targeted for LLM evidence-card generation	200	TOP_N
LLM cards generated	Article cards available for synthesis	200	Step16 article_json
Cards used for writing	Cards selected for this article package	178	Step18.1 selected cards
Included	Cards with final screening decision = include	178	Step16 final_decision
Excluded	Cards with final screening decision = exclude	19	Step16 final_decision
Uncertain	Cards with final screening decision = uncertain	3	Step16 final_decision
Full text available	Cards processed with full text available	198	Step16 has_fulltext
Abstract/metadata only	Cards processed without full text	2	Step16 has_fulltext

Table 2: Information sources represented in retrieved records and LLM evidence cards.

Information source	Retrieved records	Evidence cards
Web of Science	0	0
IEEE Xplore Digital Library	0	0
ACM Digital Library	0	0
ScienceDirect (ELSEVIER)	0	0
SpringerLink	0	0
Google Scholar	0	0
Semantic Scholar	707	115
Cornell University	0	0
Computing Research Repository	0	0
Database Systems and Logic Programming (DBLP)	0	0
ScienceDirect	0	0
DBLP	0	0
OpenAlex	1403	425
Crossref	1525	251

Information source	Retrieved records	Evidence cards
Europe PMC	423	15
PubMed	317	8
IEEE Xplore	0	0
Scopus	0	0

Table 3: Publication year distribution among selected evidence cards.

Publication year	Evidence cards
2018	16
2019	30
2020	131
2021	1

Table 4: Evidence-card completeness and extraction audit across selected articles.

Evidence item	Cards with item	Coverage
DOI present	178	100.0%
Full text available	176	98.9%
Article type extracted	178	100.0%
Aim/purpose extracted	178	100.0%
Domain/context extracted	178	100.0%
Methods/tools extracted	175	98.3%
Datasets/data sources extracted	143	80.3%
Outcomes/targets extracted	173	97.2%
Evaluation metrics extracted	147	82.6%
Main findings extracted	178	100.0%
Limitations/future work extracted	170	95.5%
Quality evidence extracted	158	88.8%

4 Main findings

4.1 Overview of canonical themes

Eight canonical themes were identified from the included studies, with the understanding that any single study may contribute to more than one theme. The distribution of article–theme assignments, shown in Table 5, highlights the dominant attention given to dataset benchmarking, spatial feature detection, and adversarial robustness, while physiological signal–based detection remains the smallest cluster.

Table 5: Summary of canonical themes with number of assigned articles and key focus.

Theme	Articles	Key Focus
Dataset Development and Benchmarking	81	Large-scale benchmarks, cross-dataset evalua
Spatial Feature-Based Detection	55	Per-frame artifacts, warping, CNN trace
Adversarial Robustness and Anti-Forensics	40	Vulnerability, adversarial perturbations, defe
Temporal and Audio-Visual Inconsistency Detection	26	Inter-frame dynamics, eye blinking, optical
Transfer Learning and Generalization	25	Domain adaptation, data augmentation, cross-
Frequency and Transform Domain Methods	15	Wavelet, DCT, EM-based spectral analys
Ensemble and Hybrid Detection Approaches	7	Multimodal fusion, stacked models
Physiological Signal-Based Detection	3	rPPG, heart rate, biological signals

4.2 Theme-level evidence patterns

Spatial Feature-Based Detection. A consistent pattern across the 55 studies in this theme is the reliance on spatial artifacts inherent in the generation pipeline — warping, upsampling, and convolutional fingerprints — as detectable signals. Early work demonstrated that affine warping artifacts alone could distinguish real from fake images without requiring negative training samples Li and Lyu [2018]. Later approaches employed expectation-maximization (EM) algorithms to model convolutional traces left by GANs, achieving over 98% accuracy on FACEAPP-generated deepfakes but revealing limited generalization to diverse manipulations Guarnera et al. [2020b,a]. A recurring contrast is between methods that target simple warping artifacts (resource-light but potentially less discriminative for modern forgeries) and those that model complex convolutional traces (more robust but still tied to specific GAN architectures). The release of Celeb-DF demonstrated that many existing methods perform poorly on higher-quality forgeries, underscoring the need for more robust detectors Li et al. [2020]. Evaluation trends show a progression from high intra-dataset accuracy to explicit cross-dataset reporting, yet a substantial gap remains between laboratory performance and real-world applicability Tolosana et al. [2020], Nguyen et al. [2020].

Temporal and Audio-Visual Inconsistency Detection. Twenty-six studies leveraged temporal dynamics — eye blinking, optical flow, inter-frame differences — to detect deepfakes. Early work employed recurrent architectures to capture eye-blinking patterns, using CNN–RNN combinations to model temporal regularities Li et al. [2018], and Wubet* [2020]. Subsequent research proposed spatio-temporal features that capture local disruptions in correlation, enhancing generalization across manipulation types Ganiyusufoglu et al. [2020]. A contrast exists between methods that rely on a single temporal cue (e.g., eye blinking) and multimodal approaches that combine edge maps, optical flow, and audio-visual dissonance Chintha et al. [2020a]. A recurring finding is that temporal methods can achieve high accuracy on intra-dataset tests but often degrade under compression or on unseen generation techniques, mirroring the generalization challenge seen in spatial feature–based work.

Frequency and Transform Domain Methods. Fifteen studies analyzed spectral or transform domain representations — Haar wavelet, DCT, or EM-based convolutional traces — to expose generation artifacts. Methods using Haar wavelet transforms achieved detection rates above 90% by exploiting resolution and blur inconsistencies Younus and Hasan [2020], Ranjan et al. [2020]. EM-based approaches extracted fingerprints from the convolutional generation process, providing explainability by linking features to specific GAN architectures Guarnera et al. [2020a,b]. A key agreement is that deepfake generation leaves detectable traces in the frequency domain, yet many methods assume specific artifact patterns (e.g., checkerboard artifacts) that are not universally

present, leading to degraded performance on unseen manipulations Osakabe et al. [2021], Wang et al. [2019]. Generalization across diverse GANs and post-processing operations remains a major gap.

Physiological Signal-Based Detection. Only three studies exploit biological signals — heart rate derived from remote photoplethysmography (rPPG) — as an implicit descriptor of authenticity. DeepFakesON-Phys achieved above 98% AUC on Celeb-DF and DFDC using a convolutional attention network Hernandez-Ortega et al. [2020]. How Do the Hearts of Deep Fakes Beat? extended detection to source identification, attaining 97.29% fake-detection accuracy on FaceForensics++ Ciftci et al. [2020]. FakeCatcher further demonstrated generalization across multiple generators and datasets, reporting accuracies between 91.07% and 99.39%, and explicitly did not depend on the source or generator of the fake content Ciftci et al. [2024]. Despite high performance, all three works note the need to test generalization to unseen generative models and real-world distortions, indicating that robustness beyond controlled datasets remains an open challenge.

Adversarial Robustness and Anti-Forensics. A consistent arms race pattern emerges: gradient-based adversarial perturbations frequently degrade forensic model accuracy by over 30% Wang et al. [2021], Yang et al. [2020a]. In contrast, defensive measures such as artificial fingerprinting show promise for robust attribution, with fingerprints persisting across transformations Yu et al. [2018, 2020]. Evaluation trends reflect a shift from static accuracy metrics toward robustness under perturbation and transferability, though many studies remain confined to white-box settings and limited detector architectures Ruiz et al. [2020], Hussain et al. [2020]. A notable gap is the limited transferability of adversarial perturbations across different detectors and the scarcity of studies on black-box attacks.

Dataset Development and Benchmarking. Foundational datasets such as FaceForensics++, Celeb-DF, and DFDC provide large-scale, diverse training and testing resources Bondi et al. [2020], Dolhansky et al. [2020], Zi et al. [2020]. A recurring finding is that intra-dataset detection accuracy is very high (often AUC ¿ 0.98), but cross-dataset performance drops substantially — up to 29% in some comparisons Bondi et al. [2020], Tolosana et al. [2021]. Transfer learning has emerged as a key strategy to mitigate dataset shift Ranjan et al. [2020]. More recent datasets like WildDeepfake present greater realism and difficulty, challenging detectors trained on virtual deepfakes Zi et al. [2020]. Temporal modeling via recurrent architectures also yields state-of-the-art results on these benchmarks Sabir et al. [2019], Chintha et al. [2020b]. The high cost of producing diverse deepfake training videos remains a limitation.

Transfer Learning and Generalization. Transfer learning consistently improves both intra-domain accuracy and cross-dataset generalization Ranjan et al. [2020], Bondi et al. [2020]. Data augmentation techniques — including residual noise and mixed-sample schemes — further improve robustness and mitigate overfitting Luo et al. [2019], El Rai et al. [2020]. However, a substantial drop in performance persists when detectors encounter unseen manipulations or datasets. Intra-dataset AUC often exceeds 0.98, while cross-dataset AUC can drop by 20–30%, highlighting the domain shift problem Bondi et al. [2020]. Recent studies increasingly report cross-dataset evaluation metrics (e.g., AUC on Celeb-DF vs. DFDC) to measure generalization, shifting from single-dataset benchmarks.

Ensemble and Hybrid Detection Approaches. Seven studies combined multiple models, modalities, or feature types to improve detection accuracy and robustness. Methods include separable CNN with image segmentation, multimodal fusion of spatial, spectral, and temporal content, and ensemble stacking of multiple CNN backbones Lewis et al. [2020], Yu et al. [2019], Rhee [2020]. One hybrid approach achieved 61.95% accuracy on the DFDC dataset, underscoring the difficulty of the task Lewis et al. [2020]. A recurring limitation is that ensemble and hybrid

methods often require increased computational resources and may still struggle with generalization. The limited number of studies in this theme indicates that hybrid approaches are still a nascent direction in deepfake detection.

4.3 Methods, tools, and data sources

Across all themes, deep learning — particularly Convolutional Neural Networks (CNNs) and recurrent architectures (LSTM, GRU) — forms the methodological backbone. Tools and frameworks include PyTorch, TensorFlow, XceptionNet, EfficientNet, ResNet, and capsule networks. Frequency-domain methods additionally use Haar wavelet transforms, Discrete Fourier Transform, and expectation-maximization algorithms. Physiological approaches employ convolutional attention networks for rPPG estimation. Adversarial studies rely on gradient-based perturbation methods and generative adversarial networks.

The most commonly used datasets are FaceForensics++, Celeb-DF, DFDC (including the preview), and WildDeepfake. Additional datasets include UADFV, DeepFakeTIMIT, FaceForensics (without ++), and custom high-quality deepfake corpora. Several studies also use synthetic data from GANs (ProGAN, StyleGAN, StarGAN). The DFDC dataset, with over 100,000 clips from 3,426 paid actors, is the largest publicly available resource for face swap detection.

4.4 Outcomes, metrics, and evaluation practices

The predominant metric is Accuracy (often binary classification accuracy) and Area Under the Curve (AUC) of the receiver operating characteristic (ROC) curve. Other metrics include Equal Error Rate (EER), precision, recall, F1-score, and detection rate. Evaluation practices are shifting from single-dataset, intra-dataset reporting to cross-dataset benchmarks: many recent studies report performance on multiple datasets (e.g., training on FaceForensics++, testing on Celeb-DF or DFDC) to measure generalization Bondi et al. [2020], Ranjan et al. [2020]. Robustness testing against adversarial perturbations, compression, and post-processing is increasingly common, especially in the adversarial robustness theme. However, a significant portion of studies still evaluate only on synthetic, controlled datasets, limiting the ecological validity of reported performance.

4.5 Cross-cutting limitations and open problems

Several limitations recur across themes. The most critical is the generalization gap: models achieve high accuracy on datasets they are trained on but suffer substantial performance drops when applied to new manipulation techniques, higher-quality forgeries, or "in-the-wild" scenarios Bondi et al. [2020], Li et al. [2020], Tolosana et al. [2021]. Adversarial vulnerability is another cross-cutting weakness: even robust detectors can be fooled by carefully crafted perturbations, and defensive measures lag behind attack sophistication Hussain et al. [2020], Yang et al. [2020a]. The field lacks a truly generic detector that maintains high accuracy across diverse generation methods, dataset shifts, and real-world distortions.

Scalability and computational efficiency are also open problems. Many state-of-the-art methods require large training sets and heavy computational resources, limiting deployment on edge devices or in real-time applications. The high cost of producing diverse, high-quality deepfake training videos further constrains dataset development Dolhansky et al. [2020]. Finally, explainability and interpretability remain underexplored: most detection methods provide binary outputs without revealing which artifacts or features drive the decision, hindering trust and forensic applicability Guarnera et al. [2020a,b]. Addressing these limitations will require continued progress in domain adaptation, adversarial defense, scalable architectures, and explainable AI.

5 Discussion

5.1 Interpretation of the evidence

The eight themes identified in this synthesis collectively paint a picture of a field that is maturing rapidly but still confronting fundamental challenges in generalization and robustness. Spatial feature-based methods, which dominate the literature with 55 assigned articles, reliably detect low-level artifacts such as face-warping inconsistencies and convolutional traces, yet their performance collapses when encountering unseen generation architectures or post-processing operations Li and Lyu [2018], Guarnera et al. [2020b], Li et al. [2020]. This pattern is echoed in frequency-domain approaches, where wavelet and EM-based fingerprints achieve high accuracy in controlled settings but degrade under compression and across different GAN families Younus and Hasan [2020], Guarnera et al. [2020a].

The temporal and audio-visual theme (26 articles) and the physiological signal theme (3 articles) point toward complementary cues that spatial-only methods miss. Temporal inconsistency methods leverage eye blinking, optical flow, or recurrent architectures to capture inter-frame anomalies that are harder to simulate in deepfakes Li et al. [2018], Guera and Delp [2018], Ganiyusufoglu et al. [2020]. Physiological detection, though nascent, demonstrates that biological signals such as heart rate can serve as robust implicit descriptors of authenticity, achieving AUC values above 0.98 on challenging datasets Hernandez-Ortega et al. [2020], Ciftci et al. [2024]. However, the small number of studies and limited testing on diverse distortions indicate that this line of work is not yet mature.

The adversarial robustness theme (40 articles) reveals an ongoing arms race: perturbations can degrade detector accuracy by over 30%, while fingerprint-based defenses offer promising attribution capabilities Wang et al. [2021]. Transfer learning and generalization (25 articles) emerge as a central strategy for mitigating dataset shift, yet cross-dataset AUC drops of 20–29% are consistently reported Bondi et al. [2020], Ranjan et al. [2020]. The dataset benchmarking theme (81 articles) underscores that intra-dataset performance is often near-perfect (AUC 0.98), but generalization to diverse manipulations remains elusive Bondi et al. [2020], Dolhansky et al. [2020], Zi et al. [2020]. The small number of studies on ensemble and hybrid detection (7 articles) suggests that multi-modal fusion and stacking remain under-explored as a path toward robustness.

5.2 Strengths of the evidence base

The synthesis draws on 178 included studies, a substantial corpus that spans multiple detection paradigms, datasets, and evaluation metrics. The availability of large-scale, increasingly realistic benchmarks such as Celeb-DF, FaceForensics++, DFDC, and WildDeepfake provides a solid foundation for reproducible comparison Li et al. [2020], Dolhansky et al. [2020], Zi et al. [2020]. The field benefits from diverse methodological strategies: from handcrafted spatial features to deep temporal models and physiological signals, offering multiple angles on the detection problem. Transfer learning and data augmentation are widely adopted to improve generalization, and the inclusion of adversarial robustness studies highlights a maturing awareness of security considerations. The evidence-card completeness and extraction audit (performed rather than formal quality assessment due to the absence of numerical quality scores across the included studies) indicated that the majority of evidence cards contained sufficient methodological detail to support synthesis.

5.3 Gaps and limitations in the literature

Several persistent gaps are evident. First, dataset shift remains the most critical limitation: models trained on one benchmark (e.g., FaceForensics++) frequently fail to generalize to others (e.g., Celeb-

DF or DFDC), and performance on wild deepfakes is dramatically lower than on controlled virtual datasets Bondi et al. [2020], Zi et al. [2020]. Second, many spatial and frequency-domain methods assume specific artifact patterns (e.g., limited resolution, checkerboard artifacts from certain GANs) that may not be present in future generation techniques, raising questions about long-term viability Li and Lyu [2018], Osakabe et al. [2021]. Third, the physiological signal theme is limited to very few studies; the robustness of heart-rate-based detection to unseen generative models, compression artifacts, and varying lighting conditions remains untested at scale Hernandez-Ortega et al. [2020], Ciftci et al. [2024]. Fourth, adversarial robustness research is concentrated largely in white-box or gray-box settings, with limited investigation of black-box transferability across diverse detector architectures Ruiz et al. [2020], Huang et al. [2019]. Finally, ensemble and hybrid approaches are represented by only seven studies, suggesting that systematic integration of multiple complementary features is under-explored despite its potential to improve generalization.

5.4 Implications for future research

Future work should prioritize domain generalization as a core objective. This entails training detectors on multiple diverse datasets, employing domain adaptation techniques, and systematically evaluating cross-dataset performance rather than reporting only intra-dataset accuracy. The development of more challenging and ecologically valid benchmarks, akin to WildDeepfake but encompassing a broader range of manipulation types and real-world distortions (compression, resizing, varying capture conditions), would help accelerate progress. The promising results from physiological signals warrant larger-scale studies that test robustness across unseen generators, different facial regions, and practical degradations. Adversarial robustness research should move toward standardized evaluation protocols that include black-box scenarios and multiple detector families. The under-explored ensemble and hybrid theme calls for systematic investigation of multi-stream fusion, stacking, and combination of spatial, temporal, frequency, and biological modalities. Finally, the field would benefit from community-wide efforts to standardize reporting metrics and data splits to facilitate fair comparison and meta-analysis.

5.5 Limitations of the automated review process

Several limitations of this review should be acknowledged. The screening process relied on an AI-assisted ranking of 1,877 records after metadata cleaning, from which the top 1,000 records were considered for full-text retrieval; this top-ranked pool may have excluded relevant studies that scored lower but contained valuable contributions. Of the 200 article cards generated, 178 were used for synthesis after excluding 19 and assigning 3 to the uncertain category; some excluded studies may have held relevant insights that were not captured. The evidence-card completeness and extraction audit was performed without formal quality scores, meaning that the depth of reporting varied across studies and could not be uniformly weighted. The automated theme discovery and synthesis relied on a large language model, which may introduce biases in theme definition or emphasize certain patterns over others. Furthermore, the review covers studies published between 2018 and 2020; given the rapid pace of this field, more recent developments (post-2020) are not reflected. Finally, the synthesis is limited to literature in English and to studies indexed in the databases searched; relevant work published in other languages or gray literature may be omitted.

6 Conclusion

6.1 Overall conclusion

This systematic literature review synthesized 112 articles published between 2018 and 2020, organized into eight thematic areas that span spatial, temporal, frequency, physiological, adversarial, ensemble, transfer-learning, and benchmarking perspectives. The review reveals a field that has made substantial progress in detection accuracy on curated datasets yet continues to struggle with generalization, robustness, and real-world deployment. A consistent finding across themes is that intra-dataset performance often exceeds 0.98 AUC, but cross-dataset evaluations reveal performance drops of 20–30% or more, underscoring dataset shift as a central challenge. The eight themes are not mutually exclusive—studies frequently span multiple approaches—and together they highlight the need for detectors that are both accurate and resilient to unseen manipulations, adversarial perturbations, and evolving generation techniques.

6.2 Most important evidence patterns

Several cross-cutting evidence patterns emerge from the synthesis. First, deepfake detection methods that achieve high accuracy on a single benchmark (e.g., FaceForensics++ or Celeb-DF) consistently lose 20–40% of that accuracy when tested on an out-of-distribution dataset, as documented across the dataset, spatial-feature, and transfer-learning themes Bondi et al. [2020], Tolosana et al. [2021], Li et al. [2020]. Second, adversarial perturbations degrade detector performance by over 30% in white-box settings, revealing a critical vulnerability that is only beginning to be addressed through fingerprint-based attribution or defense strategies Wang et al. [2021], Yang et al. [2020a], Yu et al. [2020]. Third, complementary cues—such as temporal inconsistencies, audio-visual mismatch, and physiological signals—offer detection signals that are less sensitive to spatial artifacts, but they remain underexplored and often require dedicated architectural components Hernandez-Ortega et al. [2020], Sabir et al. [2019], Ciftci et al. [2024]. Fourth, frequency- and transform-domain methods expose artifacts in spectral representations (e.g., wavelet, DCT) that are robust to some post-processing but degrade under strong compression or when artifacts are intentionally removed Younus and Hasan [2020], Osakabe et al. [2021]. Finally, ensemble and hybrid approaches combine spatial, temporal, and frequency features, yet they represent a small fraction of the literature (seven studies) and face scalability and computational cost trade-offs Yang et al. [2020b], Rhee [2020].

6.3 Practical and research implications

For practitioners deploying deepfake detectors, the evidence clearly indicates that models must be evaluated on diverse, cross-dataset protocols and under realistic post-processing conditions; relying solely on intra-dataset benchmarks can grossly overestimate real-world performance. The systematic drop in accuracy when moving from curated datasets like FaceForensics++ to more challenging corpora such as WildDeepfake or DFDC implies that continuous monitoring and periodic retraining are necessary as generation techniques evolve Zi et al. [2020], Ranjan et al. [2020]. From a research perspective, the results emphasize the need to standardize evaluation using multiple metrics (e.g., AUC, accuracy under perturbation, cross-domain AUC) and to report performance on held-out deepfake methods. The consistent vulnerability to adversarial attacks highlights the importance of incorporating robustness testing into detection pipelines. Moreover, the success of physiological-signal-based methods (three studies reporting AUC above 98% on Celeb-DF and DFDC) suggests that integrating multiple modalities—spatial, temporal, frequency, and biological—could yield more resilient detectors Hernandez-Ortega et al. [2020], Ciftci et al. [2020]. Researchers should also

prioritize the development of universal detectors that generalize beyond the training distribution, leveraging transfer learning, domain adaptation, and data augmentation Bondi et al. [2020], Akhtar et al. [2020].

6.4 Future work

Future research should address the generalization gap by constructing larger and more ecologically valid benchmarks that encompass a wider range of generation models, compression levels, real-world distortions, and adversarial inputs. The cross-dataset degradation of 20–30% observed in multiple studies calls for novel architectures that explicitly model distribution shift, such as domain-invariant representations or meta-learning. Adversarial robustness remains an open challenge; only a few works evaluate defenses against adaptive attackers, and the transferability of adversarial examples across detector families is poorly understood Ruiz et al. [2020], Hussain et al. [2020]. Combining complementary detection signals—spatial artifacts, temporal dynamics, frequency-domain anomalies, and physiological cues—into a unified, explainable framework is a promising but underexplored direction that could improve both accuracy and interpretability. Real-time detection on streaming video, robustness to post-processing pipelines, and defense against anti-forensic countermeasures are pressing practical needs that the current literature has only begun to address. Finally, future reviews should expand the temporal window beyond 2020 to capture the rapid evolution of both generation and detection methods, and should adopt standardized reporting guidelines to facilitate meta-analytic comparisons across studies.

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